

Karthic Kumar K

Senior Software Engineer | Backend Engineering | Distributed Systems | Functional Programming

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SUMMARY

Highly skilled Software Engineer with 5+ years of experience designing and delivering production-grade backend systems. Proficient in Go, Python, JavaScript, Haskell, and Java with deep expertise in debugging, code quality analysis, and software architecture. Currently mentoring engineers and driving backend excellence at H-E-B, a Fortune 500 U.S. retailer.

TECHNICAL SKILLS

- **Domains:** Supplier Management (Supply Chain), Finance Systems
- **Languages:** Go, Python, JavaScript, Haskell, Java, Rust (learning)
- **Frameworks:** Gin, FastAPI, Express.js, Servant
- **Observability:** Datadog, OpenTelemetry, Grafana
- **Database:** PostgreSQL, MySQL, MongoDB
- **Cloud:** Google Cloud Platform (Cloud SQL, Cloud Storage, GKE, Pub/Sub) and AWS (EC2, EKS)
- **AI/ML:** Retrieval-Augmented Generation (RAG), Vector Search, Prompt Engineering

WORK EXPERIENCE

G10X Pvt Ltd

Senior Software Engineer

Remote

April, 2025 - Present

- Promoted to Senior Software Engineer on the basis of strong ownership, system design contributions, and consistent delivery of high-impact backend solutions.
- Led backend development for H-E-B's large-scale Supplier Management System, collaborating with U.S. engineering teams to deliver enterprise-grade, high-availability distributed systems.
- Architected and deployed scalable microservices using Go, PostgreSQL, GCP, Docker, and GitLab CI/CD, improving deployment efficiency by 40% and maintaining 99.9% uptime.
- Designed and built an end-to-end Retrieval-Augmented Generation (RAG) system for legal document intelligence using Python and FastAPI, implementing document ingestion, chunking, and semantic retrieval pipelines with vector search to improve response relevance and reduce hallucinations.
- Architected a modular pipeline with separated ingestion, embedding, retrieval, and generation layers, and an optimized API supporting concurrent real-time queries at ~500–950 ms latency.
- Integrated OpenTelemetry for distributed tracing and observability across services, reducing mean-time-to-debug (MTTD) by 70%.
- Spearheaded internal adoption of AI-assisted development tooling (Cursor), establishing best practices around PII safety and model API cost optimization.

Associate Software Engineer

April, 2023 - April, 2025

- Built and scaled the core Supplier Management application for H-E-B, managing thousands of supplier workflows across distributed backend services.
- Developed high-performance REST APIs and event-driven services, improving system modularity, scalability, and fault tolerance.
- Refactored legacy Haskell modules to the latest LTS versions, reducing technical debt by 35% and cutting build failures by 50%.
- Redesigned Docker base images, reducing application size by 60% and improving container startup time by 2x.
- Automated recurring manual workflows through process scripting and job scheduling, saving 48+ developer hours per month.
- Maintained comprehensive unit, integration, and functional test suites with 90%+ coverage, reducing production incidents by 40%.
- Conducted SAST and DAST vulnerability scans, identifying and remediating security flaws to meet enterprise compliance standards.

Juspay Technologies

Bengaluru

Technical Product Analyst I

Sep, 2020 - April, 2023

- Contributed to backend systems for high-scale payment processing platforms using Haskell, Node.js, PostgreSQL, Docker, and AWS.
- Rewrote core JavaScript FFI functions in Haskell, improving application performance by ~80%.
- Resolved production incidents in high-stakes payment environments, maintaining system stability and minimizing downtime.

AWARDS

Received the **Pat on the Back award** at G10X for exceptional contributions to backend development and critical production support, ensuring minimal downtime and maintaining system integrity.

EDUCATION

B.E — Electronics and Instrumentation Engineering

2017 - 2021

Bannari Amman Institute of Technology | CGPA : 8.8